

Distributed Online Voltage Control with Fast PV Power Fluctuations and Imperfect Communication

Licheng Wang, Luochen Xie, Yu Yang, Youbing Zhang, *Member, IEEE*, Kai Wang, Shi-jie Cheng, *Life Fellow, IEEE*

Abstract—Coordinated Volt-Var control methods have demonstrated their techno-economic feasibility in voltage regulation of photovoltaic (PV) rich distribution systems. However, fast fluctuating PV power and imperfect communication networks may significantly challenge the effectiveness of these methods. In this paper, a revised dynamic consensus algorithm is proposed to coordinate distributed inverters for Volt-Var control in real time. With this proposed method, Var saturation and overvoltage issues which tend to occur at downstream buses of PV rich distribution systems are significantly mitigated. To quantitatively analyse the algorithm performance in imperfect communication environments, the information delivery between agents is modelled by stochastic state transition processes among finite numbers of virtual nodes so as to quantitatively depict the random time delay and packet dropout in a discrete way. On this basis, the state transition process of the whole system is further depicted by a series of row-stochastic matrices, and the ergodic theory is used to analytically derive the algorithm tracking error in an imperfect communication environment. Our proposed method can also be extended to more complex applications, where both Var compensation and PV curtailment (or EV dispatch) are available for system voltage control. Simulation results verify the superiority of our method over traditional ones.

Index Terms—consensus, distribution systems, imperfect communication, photovoltaic (PV), voltage control.

NOMENCLATURE

Sets:

$\mathcal{J}_i$ Set of buses that have connections pointing inward to Bus i .

$\mathcal{G}_i(k)$ Set of agent i 's neighbors whose messages are received by agent i within the time the k^{th} time slot.
 $\zeta_i(k)$ Set of agent i 's neighbors whose message have not been received by agent i for the last ω time steps.

Variables and Constants (related to the voltage control algorithm)

x_i^q, x_i^c The consensus state of inverters on Bus i for Var compensation and PV power curtailment respectively.
 x_{avg}^q, x_{avg}^c The average of all consensus variables.
 x_{max}^q The threshold indicates inverters' Var compensation is about to be saturated.
 V_i, V_{max} Bus i 's local voltage and the upper limit of the system voltage.
 y_i, s_i Two auxiliary variables for calculating the consensus state $x_i(k)$ of inverters on Bus i , and their initial values $y_i(0)$ and $s_i(0)$ represent the initial reactive power output and installation capacity of inverters on Bus i respectively.
 Y, S The row vectors of auxiliary variables y_i and s_i respectively.
 D_j The out-degree of Bus j in the graph of the communication network.
 α, γ Local control steps of Var compensation and PV power curtailment respectively.
 q_i^{PV}, p_i^c Reactive power absorption of inverters on Bus i ; PV power curtailment of inverters on Bus i .
 q_i^{con} Reactive power output of inverters on Bus i according to the consensus algorithm.
 $\Delta q_i^{PV}, \Delta p_i^c$ The variations of local Var output and PV power curtailment respectively.
 $\Delta \hat{q}_i$ The local Var compensation variation of agent i between its two activation events.
 $c_{pv,i}$ The installation capacity of PV inverters on Bus i .
 $m_i(k)$ The the variable (i.e., $\frac{y_i(k)}{D_i}$ or $\frac{s_i(k)}{D_i}$) of agent i that needs to be shared with neighbour agents in the step k .
 $\sigma_i(k)$ The sum of all m_i in first k steps.
 σ_i^{new} The target agent's latest received message from agent i .
 m_i^{new} The variation between two consecutive σ_i^{new} .

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Licheng Wang (email: wanglicheng@zjut.edu.cn), Luochen Xie, and Youbing Zhang (corresponding author, email: youbingzhang@zjut.edu.cn) are with the College of Information Engineering, Zhejiang University of Technology, Hangzhou 310023, China.

Yu Yang is with the College of Electrical Engineering, Zhejiang University, Hangzhou 310013, China.

K. Wang is with the College of Electrical Engineering, Qingdao University, Qingdao 266071, China (e-mail: wkwy888@163.com).

Shi-jie Cheng (e-mail: sjcheng@hust.edu.cn) is with the School of Electrical and Electronic Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China.

n	The number of buses that have PV inverters in a distribution system. Namely, the number of agents.
μ, b	The number of links in the communication network; the maximum number of consecutive packet drops in the worst case.
<i>Variables and Constants (related to the state transition model)</i>	
τ	The time resolution of the discrete state transition model.
ξ	The number of both Type I and Type II virtual nodes of a link, and it also means the number of possible time delays in the discrete model.
$v_{\xi-d+1}^{(i,j)}$	The $(\xi - d + 1)^{th}$ virtual node belonging to the Link (i, j) that connects Agent i with Agent j .
$m_{v_{\xi-d+1}^{(i,j)}}^i$	The transition factor of the state transition from Agent i to Link (i, j) 's Type I virtual node $v_{\xi-d+1}^{(i,j)}$.

I. INTRODUCTION

AS the development of large scale photovoltaic (PV) plants in load centers becomes saturated, more and more PV power are integrated into distribution systems as distributed energy resources (DERs). To deal with potential voltage violations induced by continuously increasing PV penetration, small-size PV inverters are allowed to actively participate in system voltage regulation through providing Var compensation by the IEEE 1547 standard [1].

A. Literature Review

To achieve efficient Volt-Var control, distributed PV inverters tend to operate in a coordinated way. Such multi-inverter based voltage control tasks were conventionally modelled as centralized optimization problems assuming the availability of global information [2, 3]. Furthermore, to achieve better scalability as well as improve robustness to single point failures, centralized optimal voltage control problems were decomposed into several sub-problems, which could be solved in a distributed way with only local and neighbor information [4, 5]. It is worth noting, both centralized and distributed voltage control methods mentioned above had to be iteratively solved before the converged solution can be applied on distributed PV inverters. However, considering the volatile nature of PV power, control commands figured out by above methods may lag behind the variation of PV power in real time applications.

To facilitate online implementation, for example in [6], a gradient-projection based Volt-Var control scheme was designed for coordinated voltage regulation. With this method, distributed inverters could rapidly respond to system voltage mismatch without requiring any real time communication. In [7], a dual-ascent algorithm was adopted for distributed online voltage control. Particularly, the control command of PV inverters would be immediately carried out in each iteration step, with guaranteed optimality and convergence. Other optimization based online voltage control schemes can also be seen in [8, 9], etc.

Except for optimization based schemes, consensus algorithms are also widely used to coordinate distributed inverters for voltage control. For example, reactive power output was reallocated among PV inverters by consensus algorithms in [10, 11] to promote an effective and fair use of the overall Var compensation capacity. Furthermore, consensus based schemes were extended to be applied in cases of three-phase unbalanced distribution systems such as in [12, 13]. While, early researches often relied on static consensus for coordination, where algorithms needed to be iteratively solved in a closed form. To improve real-time control performance, dynamic consensus algorithms were designed to respond to time-varying signals in each iteration step [14]. For example, a dynamic consensus algorithm was adopted in [15] to track the average of time-varying Var outputs of distributed PV inverters. Similarly in [16], inverters were coordinated by a dynamic average consensus for Var compensation sharing. Most of these researches assumed perfect communication environment.

The performance of a distributed control algorithm heavily depends on the communication network it relies on. For small-size PV inverters, they can easily gain access to wireless communication networks with low price, but at the same time, they have to face the inevitable time delays and packet dropout. To deal with random communication delay, an asynchronous distributed control algorithm was proposed in [17] for coordinated Volt-Var control. A “freezing strategy” was adopted in a distributed voltage control algorithm in [18] to counter random communication link failures. Synchronous and asynchronous voltage control methodologies were proposed and compared in [19], and a weighted autoregressive strategy was designed to improve convergence speed under random communication delays. Similarly, the impact of communication delay on control performance was discussed in [20, 21] with simulation results, and an asynchronous approach was adopted in [20] to accelerate convergence. In [22], communication delay was added into the derived small-signal model of a microgrid, and on this basis delay-dependent criteria were then developed to detect the stability of the microgrid by tracing critical eigenvalues and cluster treatment of characteristic roots.

B. Contributions and Research Gaps

The contributions of this paper can be mainly summarised into three aspects. Our innovative elements and the comparison with previous works in each aspect are demonstrated as below:

1) Algorithm design. A distributed voltage control method with merits of dynamic tracking and robustness to imperfect communication is proposed in this paper for real time voltage regulation of PV rich distribution systems. However, most of previous works only focus on either the online application performance (e.g., [7-9, 15, 16]) or the robustness to imperfect communication environment (e.g., [19-22]) of their proposed algorithms. Few researches simultaneously considered both challenges above in their algorithm design, and even fewer scholars further quantitatively derived and

analysed the impact of communication quality as well as time-varying operating conditions (e.g., fast fluctuating PV power) on algorithm performance as in this paper.

2) Communication (state transition) modelling. A virtual-node-based communication model is developed in this paper to depict the communication quality from the aspects of both random time delay and packet dropout. With the help of this communication model, state transition processes of our designed algorithm under the condition of fast fluctuating PV power and imperfect communication environment can be formulated by a series of row-stochastic matrices, which forms the basis for further convergence analysis. In most previous literatures such as [17, 19-22], only random communication delay was considered in method design.

3) Convergence analysis. The upper limit of the algorithm tracking error is analytically derived in our manuscript. In previous works, the convergence of the dynamic consensus algorithm under a perfect communication environment was proofed in [14, 23] through calculating the vector deviation between any agent's consensus state and the overall average, which is difficult to be extended to situations of imperfect communication environments. Inspired by [24], the ergodic theory is adopted in this paper to mathematically analyse the tracking error of our proposed dynamic consensus algorithm in an imperfect communication environment considering both random time delay and packet dropout. It is worth noting, [24] only provides a proof for the convergence of static consensus algorithms, and we extend the application of the ergodic theory to the convergence analysis of dynamic consensus algorithms in imperfect communication environments on the basis of our newly established communication model.

II. COORDINATED VOLTAGE CONTROL IN IMPERFECT COMMUNICATION

A. Coordinated Volt-Var Control

Volt-Var control are widely applied on distributed inverters to mitigate their local overvoltage. Once local overvoltage is detected, PV inverters are designed to absorb a certain amount of reactive power for overvoltage mitigation as below.

$$\Delta q_i^{PV}(t) = \begin{cases} \alpha[V_i(t) - V_{max}]c_{pv,i}, & V_i(t) > V_{max} \\ 0, & V_i(t) \leq V_{max} \end{cases} \quad (1)$$

$$q_i^{PV}(t + \Delta t) = q_i^{PV}(t) + \Delta q_i^{PV}(t) \quad (2)$$

where $V_i(t)$ and $q_i^{PV}(t)$ are the detected local voltage and reactive power absorption at time instant t on Bus i respectively; $c_{pv,i}$ denotes the installation capacity of inverters on Bus i ; V_{max} means the upper limit of system voltage; $\Delta q_i^{PV}(t)$ denotes the variation of reactive power at time instant t on Bus i ; α and Δt represent the control step and the time interval of local Volt-Var control respectively.

To avoid reactive power saturation as well as voltage control failure at downstream buses of PV rich distribution systems, a coordinated scheme is needed to proportionally reallocate the Var compensation burden among all distributed

inverters according to their installation capacities. The control target is correspondingly defined as

$$x_i^q(k) \rightarrow x_{avg}^q(k) := \frac{\sum_{i=1}^n q_i^{PV}(k)}{\sum_{i=1}^n c_{pv,i}} \quad (3)$$

$$q_i^{con}(k) = x_i^q(k)c_{pv,i} \quad (4)$$

where $x_i^q(k)$ and $x_{avg}^q(k)$ represent the consensus value of inverters on Bus i and the average of all consensus variables in k^{th} step respectively; $q_i^{con}(k)$ means the reactive power output according to the consensus algorithm; n is the number of buses that have PV inverters in a distribution system.

With fast fluctuating PV power, the required Var compensation for overvoltage mitigation at different buses is time varying. Consequently, the control target of (3) needs to be tracked in a dynamic way. By consecutively inputting the variations of Var compensation of inverters, the adopted dynamic consensus algorithm is able to output consensus values $x_i^q(k)$ in each iteration step for immediate execution, and $x_i^q(k)$ will asymptotically converge to its target $x_{avg}^q(k)$ as iteration goes. Considering a communication network with bidirectional links, the iteration rule of the adopted dynamic consensus algorithm is as below:

$$y_i(k+1) = \sum_{j \in \mathcal{J}_i \cup \{i\}} \frac{y_j(k)}{\mathcal{D}_j} + \Delta q_i(k+1) \quad (5)$$

$$s_i(k+1) = \sum_{j \in \mathcal{J}_i \cup \{i\}} \frac{s_j(k)}{\mathcal{D}_j}, k \geq 0 \quad (6)$$

$$x_i^q(k+1) = \frac{y_i(k+1)}{s_i(k+1)}, k \geq 0 \quad (7)$$

where both $y_i(k)$ and $s_i(k)$ are auxiliary variables (i.e., intermediate variables) for calculating the consensus state $x_i^q(k)$, and their initial values, namely $y_i(0)$ and $s_i(0)$ are initial reactive power output and the total installation capacity of inverters on Bus i respectively; $\Delta q_i(k+1)$ represents the change of local Var output caused by PV power fluctuations in step $k+1$; $\mathcal{D}_j$ has its specific name called out-degree in Graph Theory, which means the number of connections that originate from Bus j and point outward to other buses; $\mathcal{J}_i$ denotes the set of buses that have connections pointing inward to Bus i . With updated values, $\frac{y_i(k+1)}{\mathcal{D}_i}$ and $\frac{s_i(k+1)}{\mathcal{D}_i}$ will be broadcasted to all its neighbor agents $j \in \mathcal{J}_i$.

The convergence analysis of the dynamic consensus algorithm as given by (5)~(7) with a synchronous iteration rule under perfect communication environment has been demonstrated in [14, 23]. In this paper, an asynchronous iteration rule and a checking-compensation mechanism are designed and embedded into a revised dynamic consensus algorithm to counter stochastic time delays and packet dropout in imperfect communication networks.

B. Time Delay and Asynchronous Iterations

The information delivery time between neighbour agents is not fixed [25]. To deal with the asynchrony of information exchange, an asynchronous rule is designed to wake agents up for local state updating. Specifically, local state updating will be activated in two scenarios: 1) agent i receives messages

from its neighbour j within the k^{th} time step, and these neighbours compose the set $\mathcal{G}_i(k)$; 2) agent i has not received any message from its neighbour j for consecutive ω time steps, and these neighbours compose the set $\zeta_i(k)$.

Local state updating will be executed as in (8)~(10) once at least one of the activation conditions is satisfied. Considering y_i and s_i are delivered in the same data package, they will be received, delayed or lost simultaneously.

$$y_i(k+1) = \sum_{j \in \mathcal{G}_i \cup \{i\}} \frac{y_j(k)}{d_j} + \Delta \hat{q}_i(k+1) \quad (8)$$

$$s_i(k+1) = \sum_{j \in \mathcal{G}_i \cup \{i\}} \frac{s_j(k)}{d_j} \quad (9)$$

$$x_i^q(k+1) = \frac{y_i(k)}{s_i(k)} \quad (10)$$

where $\Delta \hat{q}_i(k+1)$ in (8) represents the local Var compensation variation of agent i since its last activation event. With updated values, $\frac{y_i(k+1)}{d_i}$ and $\frac{s_i(k+1)}{d_i}$ will be shared with its neighbor agents $j \in \mathcal{G}_i \cup \zeta_i$.

C. Packet Dropout and Checking-Compensation Mechanism

Network congestion caused route buffer overflows may lead to packet dropout [26]. In this section, a checking-compensation mechanism is developed to eliminate the steady state error caused by random packet dropout in imperfect communication environment. Instead of transmitting variables to neighbour agents directly, variables will be firstly accumulated at the sending agents before delivery,

$$\sigma_i(k) = \sigma_i(k-1) + m_i(k) \quad (11)$$

where $m_i(k)$ represents the variable (i.e., $\frac{y_i(k)}{d_i}$ or $\frac{s_i(k)}{d_i}$) of agent i that needs to be shared with neighbour agents in step k ; $\sigma_i(k)$ denotes the sum of all m_i in first k steps, i.e., $\sigma_i(k) = \sum_1^k m_i(k)$, and $\sigma_i(k)$ instead of $m_i(k)$ will be broadcasted to neighbour agents in step k . As a result, the target agent (e.g., agent j) will receive a series of σ_i during the iteration process. The designed checking-compensation mechanism requires agents to first “check” the timeliness of each newly received packet σ_i^{new} through its timestamp. If the timestamp of the newly received packet is older than existing ones, this packet will be discarded. Otherwise, operations as in (12)~(13) will be executed to abstract m_i^{new} from σ_i^{new} for local state updating of agent j .

$$m_i^{\text{new}} = \sigma_i^{\text{new}} - \rho_j^{(i)} \quad (12)$$

$$\rho_j^{(i)} = \sigma_i^{\text{new}} \quad (13)$$

It is worth noting, the timestamp of received sequence σ_i^{new} may not consecutive since packet dropout may occur in some cases. However, the lost message can always be “compensated” by its following packets according to the designed mechanism.

D. Convergence Analysis

A revised dynamic consensus algorithm as in (8)~(13) is established for real-time voltage control in this paper. Following this method, each PV inverter only needs trivial

algebraic operations to achieve a coordinated voltage control in field applications. On the basis of the row-stochastic state transition matrix which will be established in Section III, the tracking error of the proposed algorithm in imperfect communication environment can be mathematically derived through exploiting the ergodic theory. Detailed derivation will be demonstrated in Section IV of this paper. In this section, the tracking error is directly provided as below:

Proposition 1: An n -agent system is strongly connected by μ communication links. Without loss of generality, all communication links are assumed to be the same in terms of time delays and packet dropout. Assuming: 1) the longest route between any two agents is u , 2) the longest time delay in any communication links is $\xi\tau$, 3) packet dropout will at most consecutively occur b times, 4) the variations of local Var output between two activation events have an upper bound as $|\Delta \hat{q}_i(k)| \leq \Delta q_{i,\max}$, then it can be proofed that with the proposed algorithm, the tracking error of any consensus variable $x_i(k)$ to its target value $x^{\text{avg}}(k)$ can be quantitatively given as

$$|x_i^q(k) - x_{\text{avg}}^q(k)| \leq \underbrace{\frac{m}{\sigma^l} \frac{|\sum_i q_i(0)|}{\sum_i c_{pv,i}} \rho^{kl}}_{\text{initial error}} + \underbrace{\frac{m \sum_i \Delta q_{i,\max}}{\sigma^l \sum_i c_{pv,i}} \left(\frac{\rho^{\frac{1}{2}}}{1-\rho^{\frac{1}{2}}} + l \right)}_{\text{dynamic error}} \quad i \in 1, 2, \dots, n \quad (14)$$

where $0 < \sigma \leq 1$; $0 \leq \rho < 1$; $m = n + \xi\mu$; $l = (\xi + b + 1)u$.

As demonstrated in (14), the tracking error can be divided into two parts: 1) The first part “initial error” is induced by system initial state $|\sum_i q_i(0)|$, and its value can be asymptotically reduced to zero as iteration goes. The convergence speed of this part is related to the communication network topology and communication quality (depicted by m and l). 2) The second part “dynamic error” cannot be eliminated by iterations. Its value is determined by $\Delta q_{i,\max}$, as well as the communication network topology and communication quality. This “dynamic error” can be mitigated if the time-varying PV power changes slower or the communication environment is improved. Particularly, the dynamic algorithm will degenerate to a static one with the “dynamic error” reducing to zero if $\Delta q_{i,\max} \rightarrow 0$. According to the analysis above, the algorithm tracking error can only be reduced but not completely eliminated, and its upper limit is a trade-off between performance and cost (communication quality, communication network topology, activation rule).

E. System Voltage Control with High PV Penetration

In scenarios with very high PV penetration, system overvoltage may not be able to completely alleviated only relies on PV inverters’ Var compensation. In such situations, active power dispatch such as PV power curtailment and electrical vehicle (EV) charging power control is needed to secure system voltage stability. Without loss of generality, coordinated voltage control with both Var compensation and PV power curtailment is considered in this section. A method

can be similarly designed if EVs are involved in system voltage control through properly adjusting their charging power.

To minimize the PV power curtailment, Var compensation has a priority to be used for overvoltage mitigation. Once PV inverters' Var compensation is about to be saturated, namely the consensus state of any bus exceeds a certain threshold x_{max}^q (without loss of generality, $x_i^q > x_{max}^q$ on Bus i), inverters will curtail their PV power output if local overvoltage is detected.

$$\Delta p_i^c(t) = \begin{cases} \gamma[V_i(t) - V_{max}]c_{pv,i}, & x_i^q > x_{max}^q \\ 0, & x_i^q \leq x_{max}^q \end{cases} \quad (15)$$

$$p_i^c(t + \Delta t) = p_i^c(t) + \Delta p_i^c(t) \quad (16)$$

where $p_i^c(t)$ and $\Delta p_i^c(t)$ denote the curtailed PV power at time instant t on Bus i and its variation respectively; γ is the control step of the local voltage control.

As designed in (8)~(13), each PV inverter only needs trivial algebraic operations to achieve fair Var compensation reallocation among all inverters. Similarly, PV power curtailment can also be fairly shared by the same dynamic consensus algorithm. Correspondingly, the control target is

$$x_i^c(k) \rightarrow x_{avg}^c(k) := \frac{\sum_{i=1}^n p_i^c(k)}{\sum_{i=1}^n c_{pv,i}} \quad (17)$$

where x_i^c is the consensus variable of Bus i which represents the ratio of curtailed PV power to inverters' installation capacities; x_{avg}^c represents the average of all consensus variables. The tracking error analysis provided in Section II-D and Section IV can also be extended to the case of PV power curtailment.

It is worth noting, to avoid unnecessary PV power curtailment as well as the Var compensation, a consensus state retreat trick is designed. Specifically, the consensus state of PV power curtailment $x_i^c(k)$ of all buses will gradually reduce with a same speed if their values keep constant for longer than 10 time steps (this means local overvoltage is not detected in 10 time steps). After $x_i^c(k)$ reduced to 0 (namely no PV power is curtailed anymore), the consensus state of Var compensation $x_i^q(k)$ will start to retreat with a same method as long as there is no overvoltage risk.

III. STATE TRANSITION MODEL

Real communication environment are continuous systems with random time delays and packet dropout. To quantitatively analyse the performance of our proposed algorithm in such imperfect communication environments, a virtual node based discrete communication model is developed in this paper. Specifically, the information delivery through an imperfect communication link is modelled by stochastic state transition processes among finite numbers of virtual nodes so as to quantitatively depict the random time delay and packet dropout in a discrete way. This forms the basis for further convergence analysis of our proposed algorithm in imperfect communication environments.

A. State Transition in One Communication Link

The stochastic state transition processes in a communication link connecting Agent i and Agent j (named as Link (i, j)) is discretely modelled in this section. Without loss of generality, the longest time delay is assumed to be $\xi\tau$ with a time resolution of τ . As shown in Fig. 1, two types of virtual nodes are introduced into the discrete communication model to mimic the stochastic time delays and packet dropout in this imperfect communication link. For any single communication link, one Type II virtual node is used to model the effect of packet dropout, and the number of Type I virtual nodes depends on the longest time delay of this link as well as the time resolution (i.e., precision of the discrete model).

1) *Stochastic Time Delays.* As shown in Fig. 1 (a), $\xi - 1$ Type I virtual nodes (namely $v_1^{(i,j)}, v_2^{(i,j)}, \dots, v_{\xi-1}^{(i,j)}$) are added between Agent i and Agent j . For any Type I virtual node $v_{k \in [1, \xi-1]}^{(i,j)}$, its superscript (i, j) means this virtual node belongs to Link (i, j) , and its subscript, for example k , denotes it is the k^{th} virtual node of Link (i, j) . According to the state transition rule in this paper, a message will randomly jump to Agent j or any Type I virtual node on Link (i, j) from Agent i in its first time slot τ (time resolution), and then it will keep jumping to its neighbor nodes one by one in each time slot τ until it arrives at Agent j . As a result, there are ξ kinds of different state transition processes, which corresponds to ξ kinds of different time delays, namely $\tau, 2\tau, 3\tau, \dots, \xi\tau$. For example, to simulate a time delay $d\tau$ ($d \in [1, \xi]$), it will require d jumps for a message delivering from Agent i to Agent j . Specifically, this message will directly move to the Type I virtual node $v_{\xi-d+1}^{(i,j)}$ from Agent i in its first jump, and then it will keep jumping to its neighbour nodes one by one until it arrives at Agent j .

2) *Packet Dropout.* The Type II virtual node in our established discrete communication model is used to temporarily store the lost message when packet dropout occurs. If a data packet is lost during delivery, the corresponding message will directly jump to the Type II virtual node from Agent i , as shown in Fig. 1 (b). Simultaneously, packets that remain in the communication link will jump to their neighbour nodes in the direction of Agent j .

3) *Information Compensation.* As designed in our checking-compensation mechanism, lost messages will be compensated by their following packets. Correspondingly, in our discrete communication model, the newly sent message together with previous lost messages stored in the Type II virtual node will be simultaneously transferred to a Type I virtual node once the communication recovers, as shown in Fig. 1 (c). It is worth noting, this newly sent message will also compensate messages which are earlier sent but later arrived than it.

4) *Inactive State.* If the activation condition as designed in Section II-B is not met, new data packet will not enter into the communication link. Messages that are already in the communication link will keep jumping forward until they arrive at Agent j .

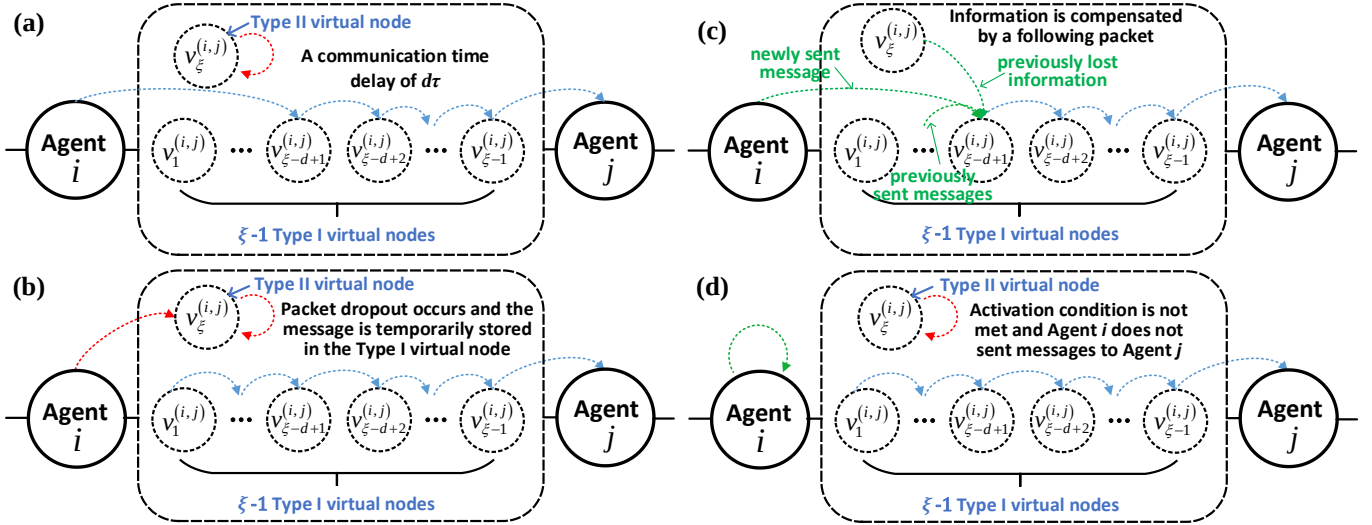


Fig. 1. The random information exchange process in an imperfect communication link with largest time delay of $\xi\tau$. (a) time delay of $d\tau$; (b) packet dropout; (c) message compensation (d) inactive state.

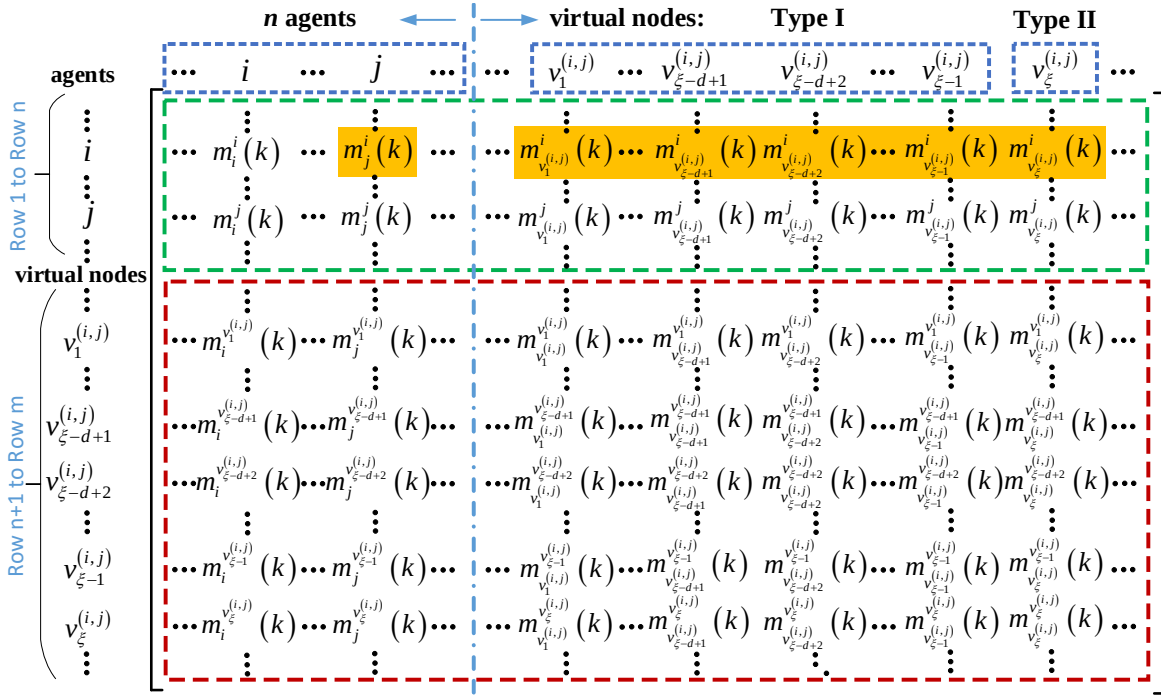


Fig. 2. State transition matrix of an n -agent system.

B. State Transition in an N -Agent System

Based on the one link model, a discrete communication model of an n -agent system is correspondingly developed in this section. Without loss of generality, the studied n -agent system is strongly connected by μ identical communication links, and each link has a maximum time delay of $\xi\tau$. Therefore, there are n agents as well as $(\xi-1)\mu$ virtual nodes of Type I and μ virtual nodes of Type II in the established model. Following the proposed algorithm, the stochastic state transition process in imperfect communication environment can be expressed as

$$Y(k) = Y(k-1)M(k-1) + \Delta Q(k) \quad (18)$$

$$S(k) = S(k-1)M(k-1) \quad (19)$$

where $Y(k), S(k) \in \mathbb{R}^{1 \times m}$, ($m = n + \xi\mu$) represent the row vectors of auxiliary variables $y_i(k)$ and $s_i(k)$ respectively, with first n items corresponding to agents (inverters) and last $\xi\mu$ items corresponding to virtual nodes. Therefore, $y_i(0) = q_i(0)$, $s_i(0) = c_{pv,i}$ if $i \in [1, n]$ and $y_i(0) = s_i(0) = 0$ if $i \in [n+1, m]$; The $1 \times m$ row vector $\Delta Q(k) = [\Delta q_1(k) \cdots \Delta q_n(k) \mathbf{0}_{1 \times \xi\mu}]$ denotes the set of local signal variations, and its element $\Delta q_i(k) = \Delta \hat{q}_i(k)$ if Agent i is activated in the k^{th} time step, otherwise $\Delta q_i(k) = 0$; $M(k) \in \mathbb{R}^{m \times m}$ represents the time-varying state transition matrix of an n -agent μ -link system in the k^{th} time step, with detailed illustration in Fig. 2. Superscript and subscript of elements in $M(k)$ denote the “from” node and “to” node respectively. For

example, $m_{v_{\xi-d+1}^{(i,j)}}^i(k)$ represents the message transition from Agent i to the Type I virtual node $v_{\xi-d+1}^{(i,j)}$ in Link (i, j) .

The following content of this section will demonstrate that the developed state transition matrix $M(k)$ is always a row-stochastic matrix (namely, a nonnegative real matrix with each row summing to 1). This is the basis to further analyse the performance of the proposed algorithm in imperfect communication environment.

1) Row 1 to Row n

Elements in Row $i \in [1, n]$ represent the state transition started from Agent i , which corresponds to the “first jump” in Fig. 1. According to the algorithm design, Agent i will send a message to Agent j through the Link (i, j) in the k^{th} step if $j \in \mathcal{G}_i(k-1) \cup \zeta_i(k-1)$. This message will arrive at either a virtual node or an agent with its first jump depending on the communication quality (time delay or packet dropout) at that time. The possible state transition of this first jump is represented by the elements in orange background as in Fig. 2, and only one of them is equal to $\frac{1}{D_i}$, with others equal to 0. While, if $j \notin \mathcal{G}_i(k-1) \cup \zeta_i(k-1)$, no message will enter into Link (i, j) in the k^{th} step, and all elements in orange background are equal to 0. Therefore, for any time step k , there are $|\mathcal{G}_i(k-1)| + |\zeta_i(k-1)|$ elements (representing the number of activated links) in Row $i \in [1, n]$ that are equal to $\frac{1}{D_i}$, with others (except for $m_i^i(k)$) equal to 0, and

$$m_i^i(k) = 1 - \frac{|\mathcal{G}_i(k-1)| + |\zeta_i(k-1)|}{D_i} \quad (20)$$

where $m_i^i(k)$ represents the self-transition; $|\mathcal{G}_i(k-1)|$ and $|\zeta_i(k-1)|$ denote the number of agents in sets $\mathcal{G}_i(k-1)$ and $\zeta_i(k-1)$ respectively.

While, if Agent i is not activated (namely all links connected to Agent i is not activated) in the $(k-1)^{th}$ step, no message will leave Agent i at time step k . In such situation, we have $m_i^i(k) = 1$ with all other elements in Row i equal to 0.

2) Row $n+1$ to Row m

Elements in Row $n+1$ to Row m represent the state transition from virtual nodes, which corresponds to the jumps after the “first jump” in Fig. 1. According to the designed discrete communication model, each virtual node will only have one target node for state transition in each time step. Namely, the out-degree of virtual nodes is always equal to 1, and from Row $n+1$ to Row m , there is only one nonzero element with a value of 1 in each row.

According to the discussion above, the sum of each row's elements is equal to 1, and the established time-varying state transition matrix $M(k)$ is always a row-stochastic matrix.

IV. TRACKING ERROR ANALYSIS

Based on the developed row-stochastic state transition matrix, the tracking error derivation of the proposed algorithm in imperfect communication environment can be divided into six steps:

Step 1: Decomposition of auxiliary variables into sub-terms.

According to the recursive iteration rule in (18), the auxiliary variable vector $Y(k)$ can be expanded as

$$Y(k) = Y(k-1)M(k-1) + \Delta Q(k) = Y(k-2)M(k-2)M(k-1) + \Delta Q(k-1)M(k-1) + \Delta Q(k) = \dots = Q(0) \prod_{i=0}^{k-1} M(i) + \sum_{r=1}^{k-1} [\Delta Q(r) \prod_{i=r}^{k-1} M(i)] + \Delta Q(k) \quad (21)$$

The right hand side of (21) is the sum of $k+1$ terms, and each term can be rewritten in a general form as

$$Q_r(k) = \begin{cases} Q(0)T_r^{k-1}, & r = 0 \\ \Delta Q(r)T_r^{k-1}, & 1 \leq r \leq k-1 \\ \Delta Q(r), & r = k \end{cases} \quad (22)$$

where $T_r^{k-1} = \prod_{i=r}^{k-1} M(i)$; $Q_r(k) \in \mathbb{R}^{1 \times m}$ denotes the impact of the increment $\Delta Q(r)$ generated at the earlier time step r (or the initial value $Q(0)$) on the state $Y(k)$ of future step k . Similarly, following the recursive iteration rule in (19), the auxiliary variable vector $S(k)$ can be expressed as

$$S(k) = S(0) \prod_{r=0}^{k-1} M(r) = S(0)T_0^{k-1} \quad (23)$$

Step 2: Normalization of sub-terms.

According to the basic matrix operation rule, for any matrix $T \in \mathbb{R}^{m \times m}$ and a row vector $X \in \mathbb{R}^{1 \times m}$, we always have

$$XT = \sum_{i=1}^m [X]_i \{T\}^i \quad (24)$$

where $[X]_i \in \mathbb{R}^{1 \times 1}$ represents the i^{th} element of the row vector X ; $\{T\}^i \in \mathbb{R}^{1 \times m}$ represents the i^{th} row of the matrix T . Therefore, (19) can be converted as

$$Q_r(k) = \begin{cases} \sum_{i=1}^m [Q(0)]_i \{T_r^{k-1}\}^i, & r = 0 \\ \sum_{i=1}^m [\Delta Q(r)]_i \{T_r^{k-1}\}^i, & 1 \leq r \leq k-1 \\ \Delta Q(r), & r = k \end{cases} \quad (25)$$

Since $M(k)$ is always a row-stochastic matrix for any time step k and the product of row-stochastic matrices will also lead to a row-stochastic matrix [27], we can conclude that T_r^{k-1} is a row-stochastic matrix. Leveraging the property of row-stochastic matrix, from (25) we have

$$\sum_{i=1}^m [Q_r(k)]_i = \begin{cases} \sum_{i=1}^m [Q(0)]_i, & r = 0 \\ \sum_{i=1}^m [\Delta Q(r)]_i, & 1 \leq r \leq k \end{cases} \quad (26)$$

On this basis, both sides of (22) can be normalized as

$$\tilde{Q}_r(k) = \begin{cases} \tilde{Q}(0)T_r^{k-1}, & r = 0 \\ \Delta \tilde{Q}(r)T_r^{k-1}, & 1 \leq r \leq k-1 \\ \Delta \tilde{Q}(r), & r = k \end{cases} \quad (27)$$

where the i^{th} element of row vectors $\tilde{Q}_r(k)$, $\tilde{Q}(0)$, $\Delta \tilde{Q}(r) \in \mathbb{R}^{1 \times m}$ are

$$[\tilde{Q}_r(k)]_i = \frac{[Q_r(k)]_i}{\sum_{i=1}^m [Q_r(k)]_i} \quad (28)$$

$$[\tilde{Q}(0)]_i = \frac{[Q(0)]_i}{\sum_{i=1}^m [Q(0)]_i} \quad (29)$$

$$[\Delta \tilde{Q}(r)]_i = \frac{[\Delta Q(r)]_i}{\sum_{i=1}^m [\Delta Q(r)]_i} \quad (30)$$

Obviously, $\sum_{i=1}^m [\tilde{Q}_r(k)]_i = \sum_{i=1}^m [\tilde{Q}(0)]_i = \sum_{i=1}^m [\Delta \tilde{Q}(r)]_i = 1$. Similar definition can also be applied on $[\tilde{S}(k)]_i$ and (23) is correspondingly normalized as

$$\tilde{S}(k) = \tilde{S}(0)T_0^{k-1} \quad (31)$$

Step 3: Difference between $[\tilde{Q}_r(k)]_i$ and $[\tilde{S}(k)]_i$.

On the basis of (27), for $0 \leq r \leq k-1$, the normalized $[\tilde{Q}_r(k)]_i$ can be expressed as

$$[\tilde{Q}_r(k)]_i = \begin{cases} \sum_{j=1}^m [\tilde{Q}(0)]_j [T_r^{k-1}]_{ji}, & r = 0 \\ \sum_{j=1}^m [\Delta\tilde{Q}(r)]_j [T_r^{k-1}]_{ji}, & 1 \leq r \leq k-1 \end{cases} \quad (32)$$

where $[T_r^{k-1}]_{ji}$ represents the element of Row j Column i in $T_r^{k-1} \in \mathbb{R}^{m \times m}$. Considering scenarios of overvoltage mitigation, $[Q(0)]_i, [\Delta Q(r)]_i \geq 0$ and consequently $[\tilde{Q}(0)]_j, [\Delta\tilde{Q}(r)]_i \in [0,1]$ always hold. Since $[\tilde{Q}_r(k)]_i$ is a linear combination of $[T_r^{k-1}]_{ji}$ with weight coefficients satisfying $[\tilde{Q}(0)]_j, [\Delta\tilde{Q}(r)]_i \in [0,1]$ and $\sum_i [\Delta\tilde{Q}(r)]_i = \sum_i [\tilde{Q}(0)]_i = 1$, we have

$$0 \leq \min_j \{ [T_r^{k-1}]_{ji} \} \leq [\tilde{Q}_r(k)]_i \leq \max_j \{ [T_r^{k-1}]_{ji} \} \leq 1 \quad (33)$$

Similar derivation can be applied on $S(k)$, and we have

$$0 \leq \min_j \{ [T_r^{k-1}]_{ji} \} \leq [\tilde{S}(k)]_i \leq \max_j \{ [T_r^{k-1}]_{ji} \} \leq 1 \quad (34)$$

Therefore, the upper limit of the difference between $[\tilde{Q}_r(k)]_i$ and $[\tilde{S}(k)]_i$ in the k^{th} step is

$$\begin{aligned} |[\tilde{Q}_r(k)]_i - [\tilde{S}(k)]_i| &\leq \max_{j_1, j_2} | [T_r^{k-1}]_{j_1 i} - [T_r^{k-1}]_{j_2 i} | \\ &\leq \max_i \max_{j_1, j_2} | [T_r^{k-1}]_{j_1 i} - [T_r^{k-1}]_{j_2 i} | \end{aligned} \quad (35)$$

Step 4: Ergodic analysis.

Two ergodic coefficients $\delta(T)$ and $\lambda(T)$ are defined as (36) and (37) to quantitatively depict the difference between any two rows of a row-stochastic matrix T [28]. The necessary and sufficient condition of $\delta(T) = 0$ is that T has identical rows, and $\delta(T) = 0$ is equivalent to $\lambda(T) = 0$ [28].

$$\delta(T) := \max_i \max_{j_1, j_2} | [T]_{j_1 i} - [T]_{j_2 i} | \quad (36)$$

$$\lambda(T) := 1 - \min_{j_1, j_2} \sum_i \min \{ [T]_{j_1 i}, [T]_{j_2 i} \} \quad (37)$$

Since all elements of a row-stochastic matrix T satisfy $0 \leq [T]_{ij} \leq 1$, we have $0 \leq \delta(T) \leq 1$ and $0 \leq \lambda(T) \leq 1$. Letting $T = T_r^{k-1}$ in (36) and substituting (37) into (35), we obtain

$$|[\tilde{Q}_r(k)]_i - [\tilde{S}(k)]_i| \leq \delta(T_r^{k-1}) \quad (38)$$

Step 5: Tracking error of each sub-term.

Proposition 2: For the parameter l defined in **Proposition 1** in Section II-D, if $k-r \geq l$, then $\delta(T_r^{k-1}) \leq \rho^{\frac{k-r}{2l}}$, where $0 \leq \rho < 1$. Namely, for a given r , $\delta(T_r^{k-1})$ asymptotically converges to 0 as k goes to infinity.

Proof: Please refer to Appendix A.

Proposition 3: For $k-l \geq 0$, the auxiliary variable $[S(k)]_i$ of any Agent i has a lower limit as

$$[S(k)]_i \geq \frac{\sigma^l}{m} \sum_{i=1}^n [S(0)]_i \quad (39)$$

where $0 < \sigma \leq 1$.

Proof: Please refer to Appendix B.

We only discuss cases of $k \geq l$ below, since l is the shortest time steps that a message can be guaranteed to arrive at any nodes of the given system as defined in **Proposition 1**. According to **Proposition 2**, (38) can be further converted to (40) for $r = 0$.

$$| \frac{[Q_0(k)]_i}{[S(k)]_i} - \frac{[S(k)]_i}{[S(0)]_i} | \leq \delta(T_0^{k-1}) \leq \rho^{\frac{k}{2l}} \quad (40)$$

Multiplying $\frac{[Q_0(k)]_i}{[S(k)]_i}$ on both sides of (40) and considering **Proposition 3** we have

$$| \frac{[Q_0(k)]_i}{[S(k)]_i} - \frac{[S(k)]_i}{[S(0)]_i} | \leq \frac{[Q_0(k)]_i}{[S(k)]_i} \delta(T_0^{k-1}) \leq \frac{m}{\sigma^l} \frac{[Q_0(k)]_i}{[S(0)]_i} \rho^{\frac{k}{2l}} \quad (41)$$

Similarly, for $1 \leq r \leq k-l$, we have

$$| \frac{[Q_r(k)]_i}{[S(k)]_i} - \frac{[S(k)]_i}{[S(0)]_i} | \leq \frac{[Q_r(k)]_i}{[S(k)]_i} \delta(T_r^{k-1}) \leq \frac{m}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \rho^{\frac{k-r}{2l}} \quad (42)$$

where $\Delta q_{i,max}$ represents the upper limit of agent i 's local Var increment $[\Delta Q(r)]_i$. Finally, for $k-l+1 \leq r \leq k$, we have

$$| \frac{[Q_r(k)]_i}{[S(k)]_i} - \frac{[S(k)]_i}{[S(0)]_i} | \leq \frac{[Q_r(k)]_i}{[S(k)]_i} \delta(T_r^{k-1}) \leq \frac{m}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \quad (43)$$

Step 6: Accumulated tracking error.

The dynamic tracking error of the proposed algorithm is defined as below

$$|x_i^q(k) - x_{avg}^q(k)| = \left| \frac{[Y(k)]_i}{[S(k)]_i} - \frac{[Q_0(k)]_i + \sum_{r=1}^k [Q_r(k)]_i}{[S(0)]_i} \right| \quad (44)$$

According to (21) and (22), we have

$$\frac{[Y(k)]_i}{[S(k)]_i} = \frac{\sum_{r=0}^k [Q_r(k)]_i}{[S(k)]_i} \quad (45)$$

Substituting (45) into (44), we have

$$\begin{aligned} |x_i^q(k) - x_{avg}^q(k)| &= \left| \frac{[Q_0(k)]_i}{[S(k)]_i} + \frac{[Q_1(k)]_i}{[S(k)]_i} + \dots + \frac{[Q_k(k)]_i}{[S(k)]_i} - \frac{[Q_0(k)]_i}{[S(0)]_i} - \frac{[Q_1(k)]_i}{[S(0)]_i} - \dots - \frac{[Q_k(k)]_i}{[S(0)]_i} \right| \\ &\leq \left| \frac{[Q_0(k)]_i}{[S(k)]_i} - \frac{[Q_0(k)]_i}{[S(0)]_i} \right| + \left| \frac{[Q_1(k)]_i}{[S(k)]_i} - \frac{[Q_1(k)]_i}{[S(0)]_i} \right| + \dots + \left| \frac{[Q_k(k)]_i}{[S(k)]_i} - \frac{[Q_k(k)]_i}{[S(0)]_i} \right| \end{aligned} \quad (46)$$

Substituting (41)~(43) into (46), we have

$$\begin{aligned} |x_i^q(k) - x_{avg}^q(k)| &\leq \frac{m}{\sigma^l} \frac{[Q_0(k)]_i}{[S(0)]_i} \rho^{\frac{k}{2l}} + \sum_{r=1}^{k-l} \frac{m}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \rho^{\frac{k-r}{2l}} + \frac{ml}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \\ &= \frac{m}{\sigma^l} \frac{[Q_0(k)]_i}{[S(0)]_i} \rho^{\frac{k}{2l}} + \frac{m}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \rho^{\frac{l}{2l}} \frac{1-\rho^{\frac{k-l}{2l}}}{1-\rho^{\frac{1}{2l}}} + \frac{ml}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \end{aligned} \quad (47)$$

Since $0 < 1 - \rho^{\frac{k-l}{2l}} \leq 1$ for $k \geq l$, the dynamic tracking error of the proposed algorithm can be finally expressed as

$$|x_i^q(k) - x_{avg}^q(k)| \leq \frac{m}{\sigma^l} \frac{[Q_0(k)]_i}{[S(0)]_i} \rho^{\frac{k}{2l}} + \frac{m}{\sigma^l} \frac{\Delta q_{i,max}}{[S(0)]_i} \left(\frac{\rho^{\frac{1}{2l}}}{1-\rho^{\frac{1}{2l}}} + l \right) \quad (48)$$

V. CASE STUDIES

A. Test System Introduction

A modified IEEE 33-bus power distribution system as shown in Fig. 3 is adopted to test the effectiveness of our proposed dynamic consensus algorithm in system voltage control. Please refer to [29] for detailed introduction of original IEEE 33-bus power distribution system and its reference. In this scenario, PV inverters and EVs can participate in system voltage control through providing Var compensation and adjusting charging power respectively. The fluctuating PV power and load profiles in a day are demonstrated in Fig. 4 and Fig. 5 respectively. The installation capacities of PV inverters and EVs are shown in Appendix C. According to the IEEE 1547 standard [1], a PV inverter will be

disconnected from a distribution system if overvoltage last longer than a certain period. Therefore, PV system owners can harvest more PV power if the system voltage can be successfully controlled within the allowable range. Simultaneously, they need to fairly take the responsibility of system voltage control through Var compensation or PV power curtailment. The voltage upper limit of the system is set to be 1.05 p.u. with a base of 12.66kV in this paper, and the active (reactive) power control step α (γ) is set to be 0.5 if voltage in kV, reactive (active) power in kVar (kW) and installation capacity in kVA. The maximum reactive power output of each PV inverter is set to be 0.4583 p.u. of its maximum active power.

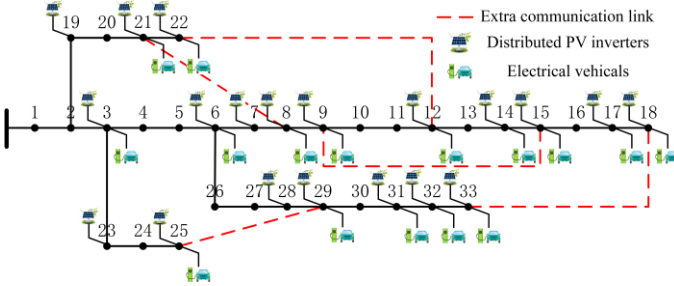


Fig. 3. A 33-bus distribution network with both PV inverters and EVs.

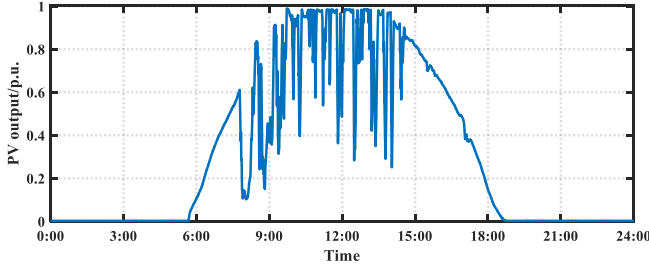


Fig. 4. A fluctuating PV power profile in one day.

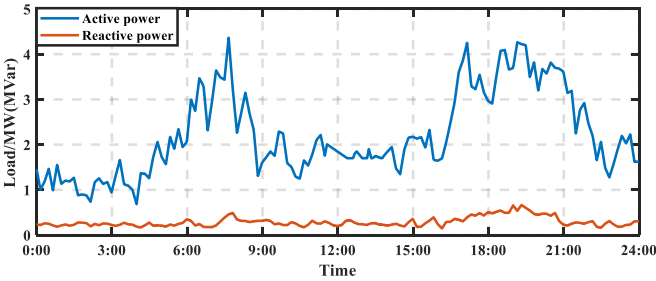


Fig. 5. Active and reactive load profiles in a day.

It is assumed to have bidirectional communication links between neighbor inverters in this distribution system, with extra links added as in Fig. 3 to facilitate algorithm convergence. It is worth noting, the communication network does not necessarily have a same topology to the power distribution system. We only need a strongly connected communication network to ensure algorithm convergence. The cumulative probability distribution of stochastic time delay of each communication link is demonstrated as in Fig. 6. The longest time delay is set to be 2s in this paper, and a data packet is regarded as lost in simulations if its time delay is larger than 2s. Correspondingly, the packet loss rate is 0.1 in

studied cases. In addition, the time resolution of the established discrete communication model is set to be 0.2s, and consequently there are at most 10 possible time delays for each communication link (namely, $\xi = 10$).

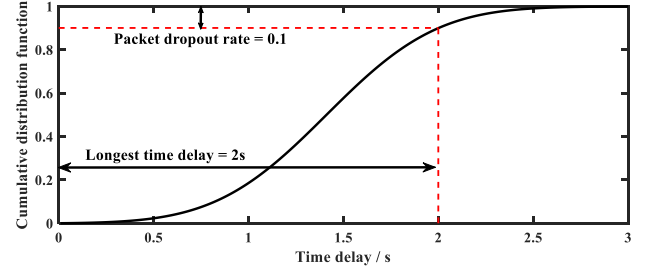


Fig. 6. Cumulative distribution function of time delay.

B. Moderate PV Penetration Scenario

In this moderate PV penetration scenario, the maximum PV power in Fig. 4 is set to be 3.6MW, and system voltage can be successfully controlled within its allowable range if PV inverters' Var compensation is properly dispatched.

1) Online Application Performance

To mitigate the Var compensation burden at downstream buses, a static consensus algorithm with a synchronous iteration rule (baseline method) is adopted to periodically reallocate Var compensation burden among all distributed inverters. The time interval of iterations is set to be 0.2 second. Considering stochastic time delays and packet dropout, agents (inverters) may reuse their last updated values to advance algorithm iteration if required signals are not received in time. However, such a signal reuse trick will not accelerate the algorithm convergence speed. With given communication network in this case study, the adopted static consensus algorithm is able to converge around 150 iterations. Namely, Var compensation burden will be reallocated among all inverters every 30 seconds.

To demonstrate the superiority of our proposed method in online application over a static consensus algorithm, the performance of Var compensation reallocation of different methods is compared in Fig. 7, where the red line representing the average Var output of all inverters in real time (namely, the tracking target). As shown in Fig. 7 (a), inverters at upstream buses (e.g., Bus 6) are able to participate in system voltage regulation with the static consensus algorithm, and these upstream inverters will adjust their Var output every 30 seconds following the algorithm commands. However, considering the volatility of PV power, such a static reallocation cannot effectively track the average Var output of all inverters in real time. Consequently, there is a very uneven Var compensation distribution among different buses and Var compensation is easy to be saturated on downstream buses. As shown in this figure, following the static consensus algorithm, inverters' Var compensation on Bus 18 first become saturated at 2:29, and it finally induces overvoltage problems on downstream buses as in Fig. 8 (a) if PV power experience a further increase.

Through consecutively injecting time-varying signals $\Delta \hat{q}_i$ into local state updating of each agent, the proposed method is

able to track the average Var output of all inverters in a dynamic way. As a result, Var compensation firstly induced at downstream buses will be rapidly diffused to upstream buses, which makes inverters at downstream buses difficult to be saturated as in Fig. 7 (b), and overvoltage at these buses can be mitigated by local Var compensation as in Fig. 8 (b). It is worth noting, the increment of local Var compensation will reduce to 0 (namely, $\Delta\hat{q}_i = 0$) if overvoltage risk is eliminated when for example PV power stops rising. In such a situation, the dynamic tracking problem is converted to a static one, and consequently the tracking error will rapidly reduce to 0.

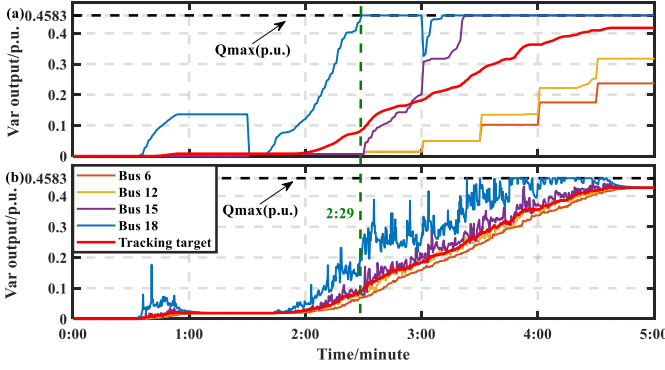


Fig. 7. Var compensation burden reallocation (a) static consensus algorithm (b) dynamic consensus algorithm.

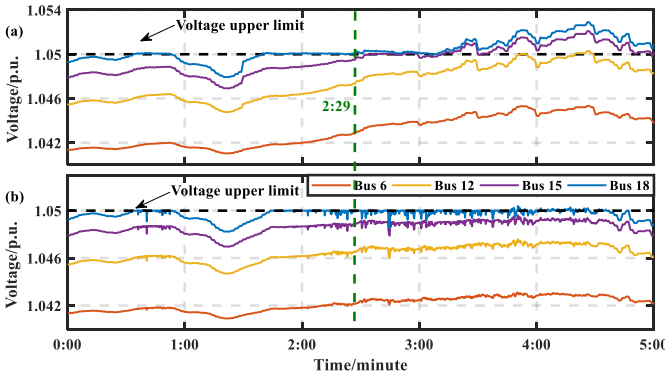


Fig. 8. Voltage profiles (a) static consensus algorithm (b) dynamic consensus algorithm.

2) Different Activation Rules

According to the design of the proposed method, agent i will send its updated local state $\frac{y_i(k+1)}{D_i}$ and $\frac{s_i(k+1)}{D_i}$ to neighbour j if agent i has not received any message from this neighbour for consecutive ω time steps. The setting of ω has impact on the communication burden. As shown in Fig. 9, compared with the synchronous iteration rule, the proposed method only needs 93.3%, 69.3% and 52.0% of communication burden if ω is set to be 1, 4 and 7 respectively. Although increasing ω can mitigate the communication burden, larger ω also means larger $\Delta q_{i,max}$, which will finally increase the tracking error of the proposed method. To balance the communication burden and tracking error, ω is set to 4 in this paper.

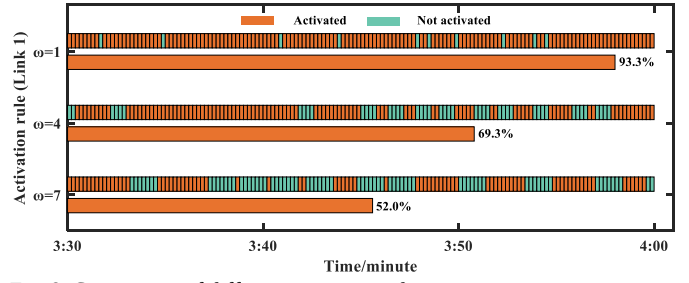


Fig. 9. Comparison of different activation rules.

3) Voltage Control Performance

The voltage control performance of our proposed method is compared with a typical Volt-Var droop control curve shown in Fig. 10. Compared with Volt-Var curve based methods that provide Var compensation depending on inverters' local voltage, our proposed method is able to sufficiently exploit the Var compensation ability of all PV inverters on both upstream and downstream buses for overvoltage mitigation. As a result, our proposed method has a better voltage control performance in distribution systems especially with high PV penetration.

The corresponding voltage profiles and Var outputs of inverters on different buses are shown in Fig. 11 (a) and (b) respectively. As shown in this figure, overvoltage occurs on downstream buses (e.g., Bus 18) after their local Var output become saturated. However, during the same period, Var outputs of inverters on upstream buses (e.g., Bus 6) remain at a low level since their local voltage is relatively healthy. In the same scenario, our proposed method can successfully control the system voltage within the allowable range as shown in Fig. 12 (a), and the required Var compensation burden is fairly shared among all inverters, as shown in Fig. 12 (b).

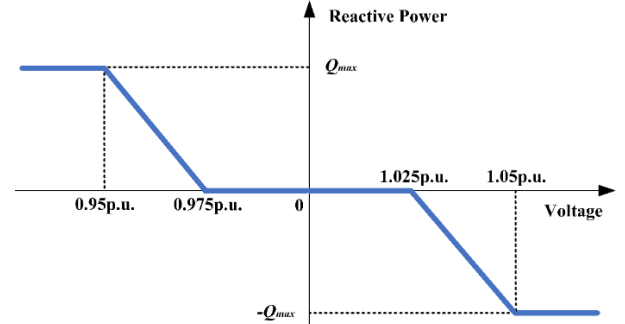


Fig. 10. A typical Volt-Var droop control curve.

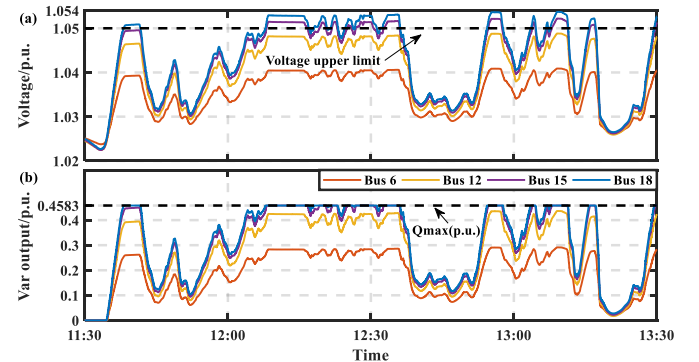


Fig. 11. Performance of a Volt-Var droop control curve (a) voltage profiles (b) Var output.

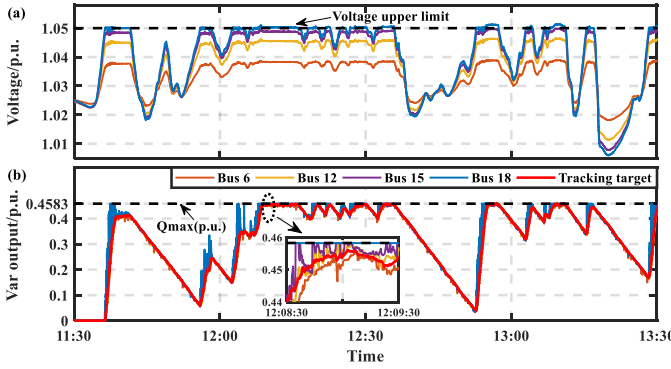


Fig. 12. Performance of our proposed method (a) voltage profiles (b) Var output.

4) Unbalanced distribution systems

With proper inter-phase coordination design in authors' previous work [15], the application of our proposed dynamic consensus algorithm can also be extended to unbalanced distribution systems for system voltage control. Please refer to [15] for detailed introduction of the unbalanced test system.

An example is demonstrated in Fig. 13 to illustrate how the proposed method works. The voltage on the Phase B of Bus 16 (the most remote bus) reaches 1.05p.u. (upper limit) at 12:02:37, and at the same time, the reactive power output of inverters on Phase B has already been saturated as in Fig. 13 (b), which triggers the inter-phase coordination control action. As designed, request of Var retreat (RVR) signals are emitted by the inverters on Phase B to the inverters on Phase C for mutual Var compensation effect (MVE) service request. Unfortunately, the local voltage of Phase C is already very close to its upper limit ($V_{local} > V_{thr}$) at this moment, and providing MVE service will further increase its local voltage. Therefore, the inverters on Phase C pass RVR signals to the inverters on Phase A. Since the voltage at Phase A is far from its upper limit at this moment, inverters on Phase A can immediately reduce its Var absorption (or increase Var injection) to provide MVE service, which will mitigate the voltage rise on Phase C and increase the voltage on Phase A. With the help of MVE from Phase A, the inverters on Phase C can further reduce its local Var absorption (namely providing MVE for Phase B) without inducing overvoltage on its own phase.

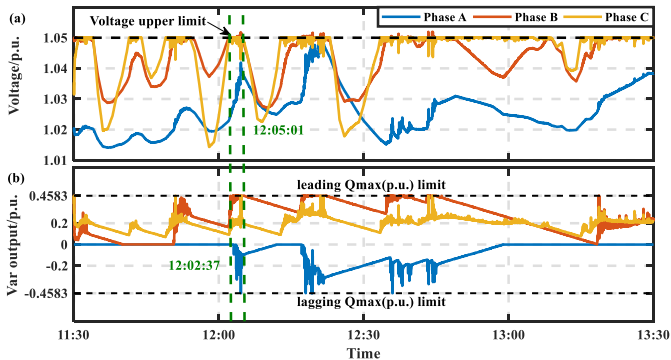


Fig. 13. Voltage regulation performance in an unbalanced distribution system (a) voltage of Bus 16 (b) Var output of Bus 16.

Different from intuitions that inverters are only designed to absorb reactive power for overvoltage mitigation, inverters on Phase A are requested to inject reactive power in this unbalanced scenario for inter-phase coordinated voltage regulation. With the state retreat trick, local consensus states of inverters are designed to gradually reduce with a same speed if their values keep constant for longer than 10 time steps (2s). As shown in Fig. B13 (b), for example after 12:05:01 the distributed inverters gradually reduce their Var output when overvoltage risk is eliminated as PV power decreases.

C. High PV Penetration Scenario

In this high PV penetration scenario, the maximum PV power in Fig. 4 is 4.5MW, and PV inverters' Var compensation alone cannot completely mitigate system overvoltage problems any more. Therefore, PV power curtailment and EV charging power adjustment are considered respectively for system voltage control after PV inverters' Var compensation is about to be saturated.

1) Voltage Profiles in Absence of Any Controller

As shown in Fig. 14, the voltage of bus 18 (the most remote bus of the PV rich distribution system) will increase to as high as 1.095p.u. if neither Var compensation nor PV power curtailment is applied to mitigate system overvoltage.

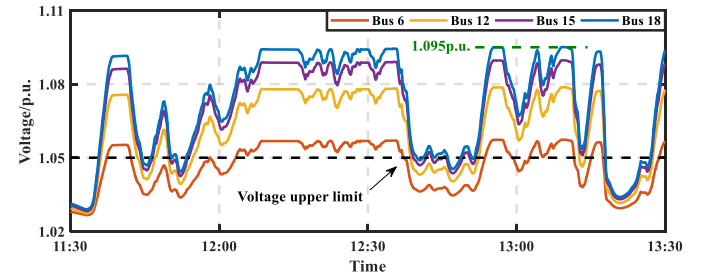


Fig. 14. The voltage profile in absence of any controller.

2) Voltage control with only Var compensation

As shown in Fig. 15, the system voltage can be successfully controlled within the allowable range before inverters' Var compensation become saturated. However, inverters lose the control of local voltage due to Var saturation after 11:35:36, and system voltage will float to higher than the safe threshold (1.05 p.u.) as PV power further increases. Without PV power curtailment, system voltage will increase to higher than 1.06p.u. in this scenario.

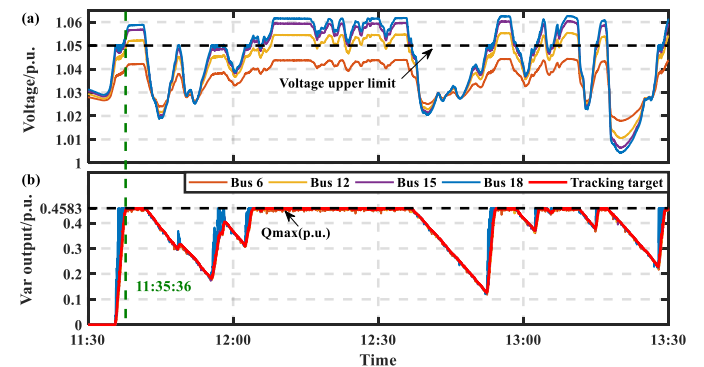


Fig. 15. Voltage control with only Var compensation (a) voltage profile (b) Var output.

3) Var Compensation and PV Power Curtailment

As designed in our proposed method, PV power curtailment will be triggered after inverters' Var compensation is about to be saturated (namely, $x_i^q > x_{max}^q$). As shown in Fig. 16, PV power curtailment is first executed on Bus 18 since the consensus x_{18}^q exceeds its threshold x_{max}^q . After that, the required PV power curtailment for overvoltage elimination is rapidly reallocated among all inverters according to their installation capacities with the help of our designed dynamic consensus algorithm. Consequently, the system voltage can be successfully controlled and the total required PV power curtailment is fairly taken by all inverters on different buses.

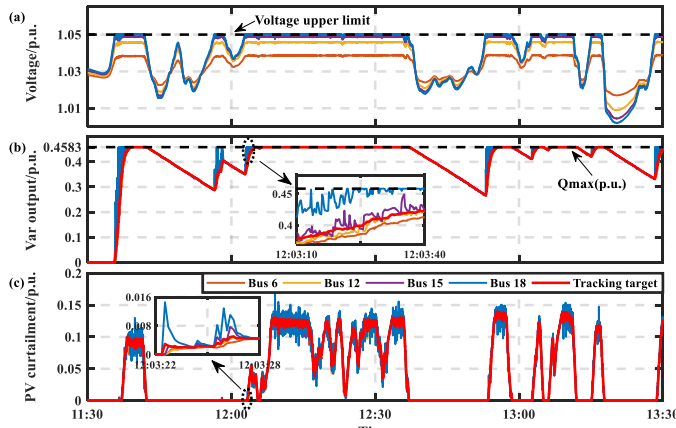


Fig. 16. Voltage control with both Var compensation and PV curtailment (a) voltage profiles (b) Var compensation (c) PV curtailment.

Furthermore, our proposed method is compared with a centralized AC optimal power flow (OPF) based method [7] in terms of harvested PV energy. It is worth noting, such an OPF method does not exist in real life since it needs global information of a distribution system and it consider neither the time delay of information delivery nor computation time cost.

According to the simulation results, the adopted AC OPF can harvest around 2.15% more PV energy in comparison to our proposed method (6.51MWh versus 6.37MWh) during the noon period (11:30~13:30). However, as shown in Fig. 17, AC OPF method tends to have more PV power curtailment on downstream buses which have larger active power-voltage sensitivity than that of upstream buses. Conversely, our proposed method aims to fairly reallocate the total required PV power curtailment among all inverters. As a result, inverters on different buses can harvest a same percentage of PV energy with our proposed method.

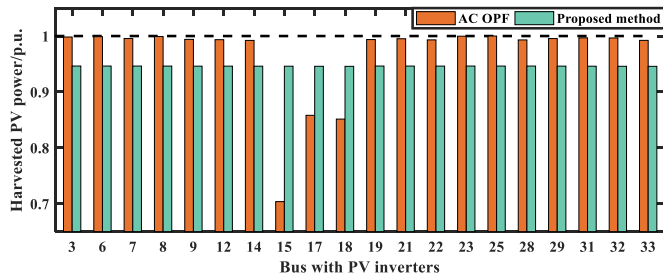


Fig. 17. Comparison of harvested PV energy of inverters on different buses.

4) Var Compensation and EV Charging Power Adjustment

The system voltage regulation performance as well as active/reactive power control are demonstrated in Fig. 18 (a), (b) and (c) respectively. As shown in Fig. 18 (a), system voltage can always be successfully controlled. According to our design, reactive power of PV inverters has a priority to be used for overvoltage suppression, and Var compensation first induced at downstream buses (e.g., Bus 18) will be rapidly diffused to upstream buses with our designed dynamic consensus algorithm. As shown in Fig. 18 (b), the Var output ratios of inverters on different nodes are very close to each other, which means the Var compensation burden is fairly taken by all inverters. Once PV inverters' Var compensation becomes saturated on any bus (e.g., $x_i^q > x_{max}^q$), EVs begin to participate in the system voltage control through properly adjusting their charging power. As shown in Fig. 18 (c), after PV inverters' Var compensation is about to be saturated, EVs start to increase their charging power for overvoltage mitigation. With the help of both active and reactive power control from EVs and PV inverters respectively, the system voltage is successfully regulated within its allowable range.

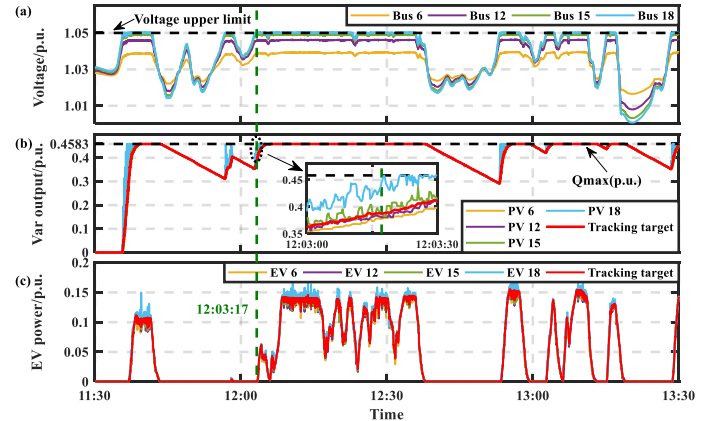


Fig. 18. Voltage control with Var compensation and EV charging power adjustment (a) voltage profiles (b) Var compensation (c) EV charging power adjustment.

VI. CONCLUSION

A modified dynamic consensus algorithm is proposed in this paper to coordinate distributed inverters for online Volt-Var control with time-varying PV power and imperfect communication environment. Based on the established discrete communication model and the ergodic theory, the algorithm performance is quantitatively analysed.

Our proposed method has been compared with a static consensus based method and a Volt-Var curve based method to verify its superior online application performance and voltage regulation performance respectively. With designed inter-phase coordination method, our proposed method can also successfully control the system voltage in a very unbalanced distribution system. The application of our proposed method can further be extended to more complex systems where both Var compensation and PV power curtailment (or electrical vehicles) are able to be dispatched for

system voltage control. Various simulation results sufficiently verify the effectiveness of our proposed method in different scenarios.

VII. APPENDIX

A. Proof of Proposition 2

For $k - r \geq l$, we define the product of l consecutive state transition matrices from $M(r + (h - 1)l)$ to $M(r + hl - 1)$ as

$$W(h) = T_{r+(h-1)l}^{r+hl-1} = \prod_{i=r+(h-1)l}^{r+hl-1} M(i), h \geq 1 \quad (A1)$$

Then T_r^{k-1} can be rewritten as

$$T_r^{k-1} = W(1) \cdots W(h)M(r + hl) \cdots M(r + hl + \Delta) \quad (A2)$$

where $h = \lfloor \frac{k-r}{l} \rfloor$ means the largest integer not greater than $\frac{k-r}{l}$, $\Delta = k - 1 - r - hl$.

Lemma 1: For parameters l and n defined in **Proposition 1**, elements of the first n columns of the product of l state transition matrices $T_{k-l}^{k-1} = M(k-l) \cdots M(k-1)$ are always positive.

Proof: According to the definition of the state transition matrix, $[M(k)]_{j_1 j_2} > 0$ if a message is transmitted from node j_1 to node j_2 in step k ; otherwise $[M(k)]_{j_1 j_2} = 0$. Since elements of $M(k)$ are always nonnegative, following the matrix multiplication rule, $[M(k)]_{j_1 j_2} > 0$ and $[M(k+1)]_{j_2 j_3} > 0$ will lead to $[M(k)M(k+1)]_{j_1 j_3} > 0$, which represents that a message is delivered from node j_1 to node j_3 with two steps. If the node j_3 is an agent, $[M(k)M(k+1)]_{j_1 j_3} > 0$ leads to $[M(k) \cdots M(k+\hat{k})]_{j_1 j_3} > 0$ for any $\hat{k} \geq 1$, since $[M(\bullet)]_{j_3 j_3} \geq \frac{1}{\mathcal{D}_{j_3}} > 0$ always holds.

According to the definition of l in **Proposition 1**, a message can always be successfully transferred to any place of the given system within l steps. Therefore, for $\forall i, j$, there exists at least one $\hat{k} \in \{1, 2, \dots, l\}$ that makes $[T_{k-l}^{k-\hat{k}}]_{ij} > 0$ holds. Furthermore, if the node j is an agent, $[T_{k-l}^{k-1}]_{ij} > 0$ always holds. It is worth noting, the first n columns of T_{k-l}^{k-1} represents the state transition from a virtual node or agent to an agent in l steps, as shown in Fig. 2. Consequently, the first n columns of T_{k-l}^{k-1} are always positive. ■

Lemma 2 [27]: If at least one column of a stochastic matrix T is strictly positive, we have $\lambda(T) < 1$.

Lemma 3 [24]: For any series of row-stochastic matrices $M(i), \dots, M(j)$, we always have

$$\delta(\prod_{r=i}^j M(r)) \leq \prod_{r=i}^j \lambda(M(r)) \quad (A3)$$

Since the established state transition matrices are discrete, there are only finite numbers of different $W(h)$. Collect all $W(h)$ in set $\mathcal{M}$ and define value $\rho := \max_{W(h) \in \mathcal{M}} \{\lambda(W(h))\}$.

According to **Lemma 1** and **Lemma 2**, we have $0 \leq \rho < 1$. Furthermore, combining (A2) with **Lemma 3**, we have

$$\delta(T_r^{k-1}) \leq \prod_{i=1}^h \lambda(W(i)) \prod_{i=r+hl}^{r+hl+\Delta} \lambda(M(i)) \quad (A4)$$

Since $0 \leq \lambda(M(\bullet)) \leq 1$, $0 \leq \lambda(W(i)) \leq \rho$, $h = \lfloor \frac{k-r}{l} \rfloor \geq \frac{k-r}{2l}$, the upper limit of $\delta(T_r^{k-1})$ can be finally expressed as

$$\delta(T_r^{k-1}) \leq \rho^h \leq \rho^{\frac{k-r}{2l}} \quad (A5)$$

Namely, $\delta(T_r^{k-1}) \rightarrow 0$ when $k \rightarrow \infty$.

B. Proof of Proposition 3

Since the state transition processes are modelled in a discrete way, there are only finite numbers of different M for a given system. As a result, there is a lower boundary (defined as σ , $0 < \sigma \leq 1$) for all positive elements of finite numbers of state transition matrices. Since elements of M are always nonnegative, following the matrix multiplication rule, we can conclude the lower boundary of positive elements in T_{k-l}^{k-1} is not less than σ^l . Considering **Lemma 1** in Appendix A, all elements in the first n columns of T_{k-l}^{k-1} are not less than σ^l . According to the iteration rule as in (23), we have $S(k) = S(k-l)T_{k-l}^{k-1}$. For Agent $i \in [1, n]$,

$$[S(k)]_i = \sum_{j=1}^m [S(k-l)]_j [T_{k-l}^{k-1}]_{ji} \geq \sigma^l \sum_{j=1}^m [S(k-l)]_j \quad (B1)$$

Since $\sum_{i=1}^m [S(k-l)]_i = \sum_{i=1}^m [S(0)]_i = \sum_{i=1}^n [S(0)]_i$, there is at least one node $i' \in [1, m]$ that satisfies

$$[S(k-l)]_{i'} \geq \frac{1}{m} \sum_{i=1}^m [S(k-l)]_i = \frac{1}{m} \sum_{i=1}^n [S(0)]_i \quad (B2)$$

Therefore, the lower boundary of $[S(k)]_i$ can be finally expressed as

$$\begin{aligned} [S(k)]_i &\geq \sigma^l \sum_{j=1}^m [S(k-l)]_j \geq \sigma^l [S(k-l)]_{i'} \\ &\geq \frac{\sigma^l}{m} \sum_{i=1}^n [S(0)]_i \end{aligned} \quad (B3)$$

C. System Parameters

TABLE C1 The Installation Capacities of PV inverters and EVs

Bus	Capacities of Inverters (kVA)	Capacities of EVs (kVA)	Bus	Capacities of Inverters (kVA)	Capacities of EVs (kVA)
3	90	196	19	157	0
6	45	98	21	90	196
7	157	0	22	90	196
8	45	98	23	112	0
9	90	196	25	90	196
12	112	245	28	112	0
14	157	343	29	135	294
15	90	196	31	90	196
17	112	245	32	90	196
18	112	245	33	90	196

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Licheng Wang received the B. Eng. degree from Huazhong University of Science and Technology (HUST), Wuhan, China, in 2012, the M. Eng degree from the Zhejiang University, Hangzhou, China, in 2015, and Ph.D. degree from the University of Queensland, Brisbane, Australia, in 2019. Currently, he is a Tenure-Track Associate Professor in College of Information Engineering, Zhejiang University of Technology, Hangzhou, China. His research interests include power system operation and control, renewable energy integration into distribution systems, distributed algorithms, deep reinforcement learning and its application in networked systems.

Luochen Xie received the B. Eng. degree from Jinling Institute of Technology, Nanjing, China, in 2021. Currently, he is perusing his master degree in College of Information Engineering, Zhejiang University of Technology, Hangzhou, China. His research interests include power system operation and control, distributed algorithm and its applications in power systems.

Yu Yang received the B. Eng. and M. Eng. degrees in College of Information Engineering from Zhejiang University of Technology, Hangzhou, China, in 2018 and 2021, respectively. Currently, he is perusing his Ph.D. degree in College of Electrical Engineering, Zhejiang University, Hangzhou, China. His research interests include power system optimal control, real-time voltage control in PV rich distribution systems, distributed algorithm and its applications in power systems.

Youbing Zhang (M'18) received the B.S. and M.S. degrees in electrical engineering from Hunan University, China, in 1993 and 1996, respectively, and Ph.D. degree in electrical engineering from Huazhong University of Science and Technology (HUST), China, in 2003. He is currently a Professor with the College of Information Engineering, Zhejiang University of Technology, Hangzhou, China. His research interests include power demand side management, vehicle-to-grid technology, power quality monitoring and control, smart energy, and big data applications.

Kai Wang is currently an associate Professor of Electrical Engineering in Qingdao University. His research interests include microgrids, storage systems, energy internet.

Shijie Cheng (M'86, SM'87, F'11, LF'18) received the B.S. degree from Xi'an Jiaotong University, Xi'an, China, in 1967, the M.Sc. degree from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 1981, and the Ph.D. degree from the University of Calgary, Calgary, AB, Canada, in 1986, all in the electrical engineering. He has been a Professor with the School of Electrical and Electronic Engineering, HUST, Wuhan, China, since 1991. His research interests are power system control, stability analysis, application of Artificial Intelligence, and energy storage. Prof. Cheng is a Fellow of the Chinese Academy of Sciences.